



Lembar kerja

ASD. Stack 2025

Total soal: 25

Worksheet time: 13mins

Nama

Kelas

Tanggal

1. What will the stack contain after the following actions?

Push(1);
Push(2);
Push(3);
Push(4);
Pop();
Pop();
Push(5);

a) 1, 2, 3

c) 1, 2, 3, 4

b) 1, 2, 3, 5

d) 1, 2, 5

2. Convert the infix expression
(A + B) * (C + D)
to Postfix.

a) A B C * + D +

c) A B * C D * +

b) A B + C D + *

d) A B + C + D +

3. Consider the following operation performed on a stack of size 5.

Push(1);
Pop();
Push(2);
Push(3);
Pop();
Push(4);
Pop();
Pop();
Push(5);

After the completion of all operation, the no of element present on stack are

- | | |
|------|------|
| a) 1 | b) 2 |
| c) 3 | d) 4 |

4. If the sequence of operations

push(1)
push(3)
pop
push(2)
push(2)
pop
pop
pop
push(3)
pop

are performed on a stack, the order of popped out values

- | | |
|------------------|------------------|
| a) 3, 2, 2, 3, 1 | b) 3, 2, 1, 3, 2 |
| c) 2, 3, 2, 3, 1 | d) 1, 3, 2, 2, 3 |

5. What is the value of the postfix expression 6 3 2 4 + - *

- | | |
|--------|-------|
| a) -18 | b) 1 |
| c) 40 | d) 74 |

6. The postfix form of the expression $(A + B) * (C * D - E) * F / G$ is?

- a) $AB + CDE * - * F * G /$ b) $AB + CD * E - FG /**$
c) $AB + CD * E - F ** G /$ d) $AB + CD * E - * F * G /$

7. Convert the following Infix expression to Postfix form using a stack

$$x + y * z + (p * q + r) * s$$

- a) $xyz^*+pq^*r+s^*+$ b) $xyz^*+pq^*r+s+^*$
c) $xyz+^*pq^*r+s^*+$ d) none

8. You can replace any element position in the stack.

- a) True b) False

9. Evaluate the given postfix expression $7\ 8\ 2\ *\ 4\ /\ +$

- a) 7 b) 11
c) 8 d) None of the Above

10. If the sequence of operations – push (1), push (2), pop, push (1), push (2), pop, pop, pop, push (2), pop are performed on a stack, the sequence of popped out values

- a) 2,2,1,1,2 b) 2,2,1,2,2
c) 2,1,2,2,1 d) 2,1,2,2,2

11. Consider the following operations performed on a stack of size 5 : Push (a); Pop() ; Push(b); Push(c); Pop(); Push(d); Pop();Pop(); Push (e) Which of the following statements is correct?

- a) Underflow occurs (stack kosong) b) Stack operations are performed smoothly
c) Overflow occurs (stack penuh) d) None of the above

12. stack colours={"red" , "green" , "blue"}

After the operations pop(), pop(), push("black") what does stack colours look like?

- | | |
|-----------------------|-------------------------------|
| a) {"red" , "black"} | b) {"red", "green" , "black"} |
| c) {"blue" , "black"} | d) {"black" , "red"} |

13. What are Stacks Used For?

- | | |
|-------------------------------|--|
| a) Type in data from keyboard | b) To remove data |
| c) To stack information | d) To keep track of a sequence of operations |

14. When an interrupt occurs, where is the current instruction placed in the stack?

- | | |
|-------------|---------------|
| a) On top | b) The bottom |
| c) The side | d) No where |

15. User perform following operations on stack of size 5 then –

push(1);
pop();
push(2);
push(3);
pop();
push(2);
pop();
pop();
push(4);
pop();
pop();
push(5);

- | | |
|---------------------------------------|---|
| a) Overflow Occurs
(stack penuh) | b) Stack Operations will be
performed Smoothly |
| c) Underflow Occurs
(stack kosong) | d) None of these |

16. Which of the following statement(s) about stack data structure is/are NOT correct?

- | | |
|--|---|
| a) Stack data structure can be implemented using linked list | b) New node can only be added at the top of the stack |
| c) Stack is the FIFO data structure | d) The last node at the bottom of the stack has a NULL link |

17. Pada saat proses merubah notasi Infix menjadi notasi Postfix.
Pada saat membaca C, isi stack operator adalah
 $(A + B) * (C + D)$

- | | |
|------------|--------------|
| a) $(+ *$ | b) $(+) *$ |
| c) $*$ | d) $* ($ |

18. Merubah notasi Infix menjadi notasi Postfix
 $(A + B) * (C * D - E) * F / G$
Pada saat membaca E maka isi stack operator adalah :

- | | |
|--------------|-------------|
| a) $*(-$ | b) $*(*-$ |
| c) $(+)*(*-$ | d) $(+)*(-$ |

19. Menghitung notasi postfix $6\ 3\ 2\ 4\ +\ -\ *$.
Pada saat membaca operator $-$, maka stack operand berisi :

- | | |
|------------|----------|
| a) 6 3 2 4 | b) 6 3 2 |
| c) 6 3 8 | d) 6 24 |

20. Stack bersifat LIFO, artinya...

- | | |
|---|---|
| a) Benda yang <i>pertama keluar</i> ke stack akan menjadi yang <i>terakhir masuk</i> dari stack | b) Benda yang <i>terakhir masuk</i> ke stack akan menjadi yang <i>pertama keluar</i> dari stack |
|---|---|

21. Fungsi untuk mengambil item pada stack

- | | |
|----------|-----------|
| a) pop() | b) push() |
|----------|-----------|

Kunci Jawaban

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|--|--|--|
| 1. d) 1, 2, 5 | 2. b) $A B + C D + *$ | 3. a) 1 |
| 4. a) 3, 2, 2, 3, 1 | 5. a) -18 | 6. b) $AB + CD * E - FG$
/** |
| 7. a) $xyz * + pq * r + s * +$ | 8. b) False | 9. b) 11 |
| 10. a) 2,2,1,1,2 | 11. b) Stack operations
are performed
smoothly | 12. a) {"red" , "black"} |
| 13. d) To keep track of
a sequence of
operations | 14. a) On top | 15. c) Underflow (stack
Occurs kosong) |
| 16. c) Stack is the FIFO
data structure | 17. d) * (| 18. a) *(- |
| 19. c) 6 3 8 | 20. b) Benda yang
<i>terakhir masuk</i>
ke stack akan
menjadi yang
<i>pertama keluar</i>
dari stack | 21. a) pop() |
| 22. a) $s \rightarrow count == 5$ | 23. c) Rear | 24. b) Prints binary
representation
of n |
| 25. a) 6 24 | | |

